

Prompt Confusion Mechanistic Slice

First-pass readout on two promising families: **trade_size_force_large** and **activity_force_observe**. The slice uses aligned vs strong-conflict rows plus a strong-conflict-only strategy-vs-setting split.

DATE	MODEL	CAPTURE RUN	STATUS
April 2026	Qwen3-30B-A3B	09036218e3ae	Probe-only sanity pass

BOTTOM LINE

The slice is clearly linearly decodable, but the current probe setup is probably too easy. The conflict-vs-aligned result is so strong that it is more likely reading prompt-family or formatting differences than a clean conflict representation. The strategy-vs-setting split is more believable and worth following up, but still needs grouped controls before it supports a mechanistic claim.

CONFLICT SLICE	SOURCE SLICE	BEST CONFLICT LAYER	BEST SOURCE LAYER
1,944	666	L12	L8
Aligned vs strong-conflict rows	Strong-conflict only; strategy vs setting readout	Balanced acc 0.998	Balanced acc 0.906

SLICE DEFINITION

The probe slice comes from two Neon views over the same capture. The first view labels rows as **aligned** or **conflict** and contains **1,944** examples. The second keeps only strong-conflict rows with usable readout-side labels and relabels them as **strategy** or **setting** it contains **666** examples. Both runs use residual-stream last-token activations on the captured layers **0, 4, 8, ..., 44**.

Probe Readout

Conflict Vs Non-Conflict

The global readout is almost perfectly separable at every captured layer. Accuracy ranges from **0.967** at layer 0 to **0.998** at layer 12; balanced accuracy is effectively the same because the slice is balanced. That is too good to trust at face value.

Best layers.

Layer	Accuracy	Balanced	Selectivity
12	0.9979	0.9979	0.4830
24	0.9964	0.9964	0.4856
36	0.9938	0.9938	0.5324
4	0.9877	0.9877	0.5329

Conflict run: strong signal, but likely contaminated by prompt-level differences under random CV.

Strategy Vs Setting

The strong-conflict-only readout is weaker but still strong across the whole layer sweep. Balanced accuracy ranges from roughly **0.885** to **0.906** on **666** examples, peaking in the early-mid stack. This looks more plausible than the conflict split, but it still shares the same leakage risk.

Best layers.

Layer	Accuracy	Balanced	Selectivity
8	0.9054	0.9058	0.4521
36	0.9024	0.9038	0.4415
4	0.9009	0.9015	0.4475
32	0.8964	0.8974	0.4475

Source run: useful signal, but still not a clean mechanistic result without grouped evaluation.

Why The First Result Is Suspicious

MAIN CAVEAT

The current analysis uses ordinary cross-validation over rows. That means aligned and strong-conflict examples from the same prompt family can land in both train and test. A linear probe can therefore succeed by reading stable formatting or family-level artifacts rather than a clean “conflict state.” The near-perfect conflict-vs-non-conflict numbers are the signature to worry about, not celebrate.

What Looks Real	What Still Needs Control
There is definitely linearly decodable structure in the slice.	Whether the readout survives grouped CV by matched_pair_id or template family.
The strategy-vs-setting task is harder than the coarse conflict task.	Whether the signal remains once family and pressure-bucket shortcuts are blocked.
Early-mid layers are already sufficient for source readout.	Whether the best read layer is also the right write/localization layer.

Recommended Next Step

The next pass should be narrower and stricter rather than broader:

- 1. Re-run the two probes with grouped CV by **matched_pair_id**.
- 2. Break out **trade_size_force_large** and **activity_force_observe** separately.
- 3. Add a matched-pair delta analysis: compare **strong_conflict - aligned** inside the same pair.
- 4. Only after that, decide whether PCA or causal patching is worth the time.

Current status: good enough to continue, not good enough to claim a mechanism.